



Identifying some important success factors in adopting agile software development practices

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ABSTRACT

Agile software development (ASD) is an emerging approach in software engineering, initially advocated by a group of 17 software professionals who practice a set of “lightweight” methods, and share a common set of values of software development. In this paper, we advance the state-of-the-art of the research in this area by conducting a survey-based ex-post-facto study for identifying factors from the perspective of the ASD practitioners that will influence the success of projects that adopt ASD practices. In this paper, we describe a hypothetical success factors framework we developed to address our research question, the hypotheses we conjectured, the research methodology, the data analysis techniques we used to validate the hypotheses, and the results we obtained from data analysis. The study was conducted using an unprecedentedly large-scale survey-based methodology, consisting of respondents who practice ASD and who had experience practicing plan-driven software development in the past. The study indicates that nine of the 14 hypothesized factors have statistically significant relationship with “Success”. The important success factors that were found are: customer satisfaction, customer collaboration, customer commitment, decision time, corporate culture, control, personal characteristics, societal culture, and training and learning.

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1. Introduction

ASD is currently an emerging software engineering approach, constituting of a set of principles initially advocated by a group of 17 software practitioners, and now practiced by many software professionals. The principles they advocated, leading to the emergence of the ASD philosophy, are based on best practices and their previous success and failure experiences with many software development projects regarding what works and what does not work in practice. Each of these practitioners had their own different philosophies about how they approached software development. However, all of them advocated close collaboration between software development and business teams, as opposed to silo development by software teams; face-to-face communication, as opposed to over-emphasis on written documentation in projects; frequent delivery of portions of working software, as opposed to final delivery of the complete product at the end; accepting changing requirements by customers, as opposed to defining a fixed set of requirements “cast-in-stone”; and adaptive organizational capability of teams according to changing business requirements. These principles (Fowler and Highsmith, 2001; Fowler,

2002) underlie the philosophy of ASD. On the contrary, the traditional software development projects were more focused on following well-defined plans and detailed documentations. In February 2001, these advocates of the agile philosophy met together, and prepared the *Manifesto for ASD*. Following 2001, many other software practitioners were inspired by the philosophy behind ASD. It is pertinent to mention that prior to the birth of the ASD paradigm, different practitioners used to follow some of the best practices suggested by the ASD paradigm in their software development projects in an *ad hoc* manner. However, it is only after 2001 that these set of practices were formalized and were together coined the term “Agile”.

The overall goal of the paper is to improve the understanding of the emerging ASD approach using a survey-based statistical study. Specifically, we have attempted to address the following question:

What are the factors from the perspective of ASD practitioners that will influence the success of projects that want to adopt ASD practices?

We conducted an unprecedented large-scale empirical study involving 241 returned survey questionnaires. Most of the existing papers available in the literature report experiential and anecdotal information. Important pieces of related existing literature are presented in Section 2. The feedback we received while conducting

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the survey has been overwhelming. Many of the respondents, who are also experts in this area, have shown interest in the work and its results.

The rest of the paper is organized as follows: In Section 2, we review the background and motivation of this work. In Section 3, we hypothesize various success factors. In Section 4, we describe our research methodology. In Section 5, we describe the results of the statistical analysis. In Section 6, we discuss the research findings from different perspectives. Finally, in Section 7 we conclude the paper and provide some directions for future research.

2. Background and motivation

There are a number of pieces of literature reporting empirical studies relating to different aspects and perspectives of ASD projects. Detailed discussions on some of these studies exist in Abrahamsson and Koskela (2004), Baheti et al. (2002), Baskerville et al. (2003), Ceschi et al. (2005), Chong (2005), Dagnino et al. (2004), Dalcher et al. (2005), Jokela and Abrahamsson (2004), Karlstroem and Runeson (2005), Layman et al. (2004), Macias et al. (2003), Mann and Maurer (2005), Mannaro et al. (2004), Melnik and Maurer (2002), Middleton (2001), Reiffer (2002), Robinson and Sharp (2004), Rumpe and Schroder (2002), Sharp and Robinson (2004), Shine Technologies (2003), Sillitti et al. (2005), Salo and Abrahamsson (2008), Wellington et al. (2005) and Young et al. (2005).

In a recent work, Dybå and Dingsøyr (2008) published an excellent work in which they systematically reviewed all the empirical studies, including the above, that have been conducted till recently. However, none of these works are extremely similar in nature, scope, size and methodology with the study reported in this paper. Therefore, in order to avoid redundancy of discussions in the literature and to present our work contextually and briefly, we avoid presenting a review of all the above papers. Interested readers are referred to Dybå and Dingsøyr (2008) to get a detailed overview of the studies presented in the works mentioned in the last paragraph. Instead, we articulate below, in short, only some of those works that we believe align more directly than others with our work or that have implications on some of the discussions reported in this paper.

Reiffer (2002) is one of the first papers reporting survey results on agile methods. Although the nature of the survey was very simple, using only a small data space, it was important because it captured various experiences of ASD practitioners in various industry types. Reiffer found that five firms reported that their defect rates from agile projects were at par with their other projects.

Baheti et al. (2002) conducted empirical studies on distributed pair programming. They gathered empirical data in favor of pair programming in distributed teams, a very important issue in the current world of offshoring and outsourcing of IT business processes. Their study resulted in the creation of a special environment for distributed programming using dual screen projectors, and hypermedia-enhanced video streams.

Rumpe and Schroder (2002) conducted a survey on Extreme Programming (XP) projects. They found out that the continuous presence of customers was important in XP projects. They also found out that although there are a plenty of problems in XP's success, it was found to be superior to some of traditional software development approaches. However, their survey had few limitations in terms of size, application domains, and seeded biases. They called for further surveys on XP.

Shine Technologies (2003) conducted a simple survey of the global experiences with ASD. They found that companies that follow agile practices have lower costs, better productivity, better quality, and better business satisfaction.

Abrahamsson and Koskela (2004) performed a controlled case study on XP in practical settings. In their study, they got four software engineers to implement a web-based system of 7698 lines of code over a period of 820 h in eight weeks. They obtained favorable results in support of the usefulness of XP.

In a very recent study, Salo and Abrahamsson (2008) reported the use and the usefulness of XP and Scrum in 13 organizations of 8 European countries' complex embedded software development organizations. They found out that the embedded software development projects were able to successfully use the above agile methods.

Jokela and Abrahamsson (2004) attempted to study the relationship between usability engineering and Extreme Programming. They investigated in a project the extent to which Extreme Programming guides the development of usable software. Their findings reveal that Extreme Programming is poor in explicitly attending to the usability of software.

Karlstroem and Runeson (2005) reported the results of study of the feasibility of using agile methods in traditional stage-gate project management environments. Their findings show that despite there are possibilities of some initial management resistance of such an attempt, it is definitely possible to use agile methods in such environments.

Wellington et al. (2005) performed yet another study in which they reported the comparison of student experiences with both plan-driven and agile methods. They measured team cohesion and students' attachments to projects throughout. They also monitored the applications that were developed by the different teams. They found out that all the teams developed almost the same amount of functionality, the XP team's code had better metrics. The teams that used the traditional process had to write more code to implement the same functionality.

It is pertinent to mention at this point that although the studies reported in the papers mentioned in the above paragraphs do not directly focus on empirically identifying the success factors in adopting ASD practices in the same manner we do in the work reported in this paper, there exist a number of other studies relating to empirical identification of success factors in other software development contexts. Instead of being unnecessarily exhaustive, given the scope of this work, we list of the relevant works as follows: Dybå (2000a, 2003, 2005), El Eman et al. (1999), Goldenson and Herbsleb (1995), Rainer and Hall (2002), Stelzer and Mellis (1998) and Yousof and Aspinwall (1999).

Let us now summarize the motivations behind our work and our contributions. Since its formal birth, the ASD paradigm is increasingly becoming popular in the last few years. Because of the attractive claims of ASD, many organizations are switching to adopt agile practices in their software development projects. However, as this is an emerging software development philosophy, there are concrete empirical studies required to assess its different aspects of ASD. Large-scale empirical studies, having the scope and magnitude of what has been performed as part of this work, are scarce, if not unavailable in totality (which is not known to the best of our knowledge), and we are attempting to address this deficiency.

Let us assume that a software practitioner, who used to practice plan-driven software development, wants to adopt ASD practices in a project that is about to start. He/she would be interested to know – What are the factors on which his/her team should focus that would lead to the success of his/her project? There are potentially many factors to focus on. What is interesting to know are the critical success factors.

These are the issues that we have addressed in this work. Our hope is that the work would aid the future software practitioners wanting to adopt ASD practices in their projects. This affirms the motivations behind the work.

It should be clarified that we have not addressed the issue of whether ASD works is practice or not, although in some of the discussions in the sections to follow we have referred to this issue to put things in context. There were quite a few studies available in the literature that address the issue of success of ASD, and we had opted not to address this question in this paper.

3. Success factors

In this Section, we propose a hypothetical framework (*a priori* framework) qualitatively attempting to address the research question on the basis of intuition and the previously published literature on ASD. We also formulate a set of hypotheses which will be validated using the data collected from ASD practitioners in different industries.

While there can be a list of different factors that can be identified, the holistic view of what project managers consider “Success” in any software development project is worth considering. From a project manager’s viewpoint, time, cost, and quality (PMI, 2004; Schwaber and Beedle, 2002) are three key components of any project’s success. These are competitive criteria for software development project managers. Any disciplined software development activity has the primary goals of developing high quality software, with less time, and cost. We believe that these are three key attributes characterizing the success of any software development project, and most other success characteristics can be transformed into any of them. For example, productivity, customer satisfaction, and functionality can be perceived as quality criteria.

The following five criteria will be used to determine the success of (agile) software development (projects):

- Reduced delivery schedules.
- Increased return on investment (ROI).
- Increased ability to meet with the current customer requirements.
- Increased flexibility to meet with the changing customer requirements.
- Improved business processes.

These criteria form the constituents of the dependent variable (“Success”) in our research. These capture the three important dimensions of the success of any project, reduced time, reduced cost, increased quality.

So far, we have discussed what we consider as the measures of success. Let us now describe some of the potential candidates for success factors.¹

3.1. Organizational factors

3.1.1. Customer centric issues

Some of the authors who, according to our view, have directly or indirectly advocated/cited customer centric issues as a factor are Cohn and Ford (2003), Lindvall et al. (2002), McMahon (2004), Nerur et al. (2005), Oppertthausen (2003) and Turner and Boehm (2003). The three types of customer centric issues they have considered can be broadly classified as:

- Customer satisfaction
- Customer collaboration
- Customer commitment

The Agile Manifesto advocates customer collaboration as one of the important requirements for successful software development. One of the principles of ASD is giving highest priority to achieving customer satisfaction through early and continuous delivery of valuable software (Fowler and Highsmith, 2001). This requires that the customers are not only available on-site with the software development team, but also highly motivated, active, and consider themselves responsible elements in the project. Customer commitment is, thus, an important success factor.

3.1.2. Decision time

Successful ASD teams are normally left on their own, thereby allowing them to take their own decisions and succeed. Also, as close customer, business area and software developer collaborations are advocated by agile practices (Fowler and Highsmith, 2001), important project decisions are likely to be made within a short timeframe. Agile practices are perceived to likely succeed in environments where rapid communication is enabled. Rapid communication is likely to cut down the amount of time expended on major decisions. Therefore, fast decision making is prominently an enabler of agility.

3.1.3. Team distribution

Having centralized teams is also a likely factor positively affecting the success of projects. Co-located teams are often recognized as one of the important vehicles for successful communication, which is, in turn, is one of the important success factors of software development. Companies involved in distributed international projects will be affected by the cultural and political situations in those regions. The geographic distribution and the location of the teams are important because the local political, cultural, and behavioral habits and situations greatly influence the productivity of the project team. For instance, in regions of the world where people are used to working collectively, agile approach will likely to be more successful in organizations that are located there than in organizations located in regions where that is not the case.

3.1.4. Team size

The number of members in a team greatly influences the degree of communication that can be had between team members. For instance, in a team that has a large number of members, it is quite likely that frequent, informal, and rapid interactions will become difficult. The pace of informal communications can be made faster in small teams. Having a larger team might pose great hindrance to fast communication and decision making in projects. However, if there are more people in a project, it is suggested that large teams be broken down into few smaller teams. Further, with the growth of the team size, efficiently coordinating/managing the interactions between people might be an important issue to be taken care of.

3.1.5. Corporate culture

Having the right corporate culture is almost unanimously perceived by agile experts to be a necessary factor determining the introduction of agile practices (Lindvall et al., 2002). Since agile practices emphasize “individuals and interactions over processes and tools”, and “customer collaboration over contract negotiation” (Agile Manifesto), intuitively, the nature of organizations individuals work in is important. For example, ASD is not appropriate in bureaucratic organizations (Lindvall et al., 2002). A dynamic and fast changing organization will find ASD methods extremely suitable for it (Abrahamsson et al., 2002). Since the agile practices put emphasis on customer collaboration and feedback, the organizational culture should be adaptive to changes. The organizational

¹ In this work, we have focused on Organizational Factors and People Factors only.

culture should also be supportive of working in a collaborative environment.

3.1.6. Planning and control

One of the important aspects that characterize the implementation of ASD practices is the nature of organizational, management, and project planning and control. For instance, documented plans, accompanied by quantitative performance measures, are considered key to the success of organizations practicing traditional/plan-driven software development. On the contrary, internalized plans, and qualitative control prepared and monitored by the team rather than external managers are considered to succeed organizations adopting ASD practices (Boehm and Turner, 2003). Some of the other authors who have given importance to planning and control as a factor are Cockburn (2002a,b), Cohn and Ford (2003), McMahon (2004).

3.2. People factors

The success of a software development project depends on people and human resource factors. Evidently, this becomes more so in ASD. This is because in ASD practices, there is emphasis on individuals and interactions, customer collaboration, and responding to changes suggested by customers. We list below some of the important people factors. Some of the factors mentioned below could equally be classified as organizational factors. However, instead of getting bogged down into the less important issue (according to the goals of this work) of which category they belong to, we simply describe them here.

3.2.1. Competency

Competency refers to whether one has real-world experience in the technology domain, has built similar systems in the past, and possesses good interpersonal and communication skills (Lindvall et al., 2002). In the context of talent hunt, the following principle of Barry Boehm should be kept in mind: “use better and fewer people” (Cohn and Ford, 2003; Keith, 2002). Since the agile practices emphasize delivering working software fast, evidently, the competency of the team members in a project is important. From the discussions in Lindvall et al. (2002), it can be inferred that having at least 25% technically competent people in the projects is important. Previous experience in developing software similar to the current one is important in a project. The competency of the team members dictate not only the pace of development of a software, but also delivering the software according to the requirements of the customers.

3.2.2. Personal characteristics

Having right people, may not necessarily be in terms of experience, is believed to be key to success with big teams (Lindvall et al., 2002). Not only the experience and competency of the team members is important, but also their personal characteristics such as honesty, collaborative attitude, sense of responsibility, readiness to learn, and work with others are considered equally important, if not more (Lindvall et al., 2002).

3.2.3. Communication and negotiation

Communication plays an important role in the implementation of ASD practices in a software development project (Turner and Boehm, 2003). Effective communication is identified as one of the important factors for agile practices to succeed (Ambler, 2005c). One of the important requirements for agility is fast and effective communication between developers, operations, support, customers, management, and business areas.

3.2.4. Societal culture

Software development is similar to the development of other products in that the culture of the society in which the organization operates is important. As any other human activity, software development is highly influenced by regional (local) culture. The Agile Manifesto (Fowler, 2002) lays emphasis on individuals and interactions between people over processes and tools. The agile principles (Fowler and Highsmith, 2001) suggest using motivated individuals. People should be eager to learn from each other, be honest, collaborative, and responsible (Lindvall et al., 2002). All these are affected by the societal cultural factors as well. For instance, if a society has communicative people, if they are dynamic and progressive by their nature, their work habits will be affected as well. As we have emphasized earlier that personal characteristics are important success factors, so would be the societal culture, because the personal characteristics of the people are greatly influenced by their societal culture and vice versa.

3.2.5. Training and learning

An important concern that has been raised in the previous literature is how much training is required for successfully implementing ASD practices in an organization (Lindvall et al., 2002). In ASD, emphasis is on continuous learning (Lindvall et al., 2002). People should be eager to share information with one other, continuously learn. This increases the chances of ASD practices. It is perceived that ASD practices require less formal training (especially in the case of XP, because of pair programming), mentoring, and professionally guided discussions were more useful in ASD than formal training, so that tacit knowledge could be transferred between individuals. Honest and collaborative people in a project often makes training easier (Lindvall et al., 2002).

3.3. Research hypotheses

Fig. 1 summarizes the success factors we have considered. We also hypothesize the following based on the discussions in Sections 3.1 and 3.2. The truths of the hypotheses are validated using data collected from surveys.

HO1a: The greater the satisfaction of the customers in projects, the more likely would be the success of the ASD projects.

HO1b: The greater the collaboration of the customers in projects, the more likely would be the success of the ASD projects.

HO1c: The greater the commitment of the customers in projects, the more likely would be the success of the ASD projects.

HO2: The quicker the decisions are taken in projects, the more likely would be the success of the ASD projects.

HO3: The more closely located the project teams are, the more likely would be the success of the ASD projects.

HO4: The smaller the size of the teams in a project, the more likely would be the success of the ASD projects.

HO5: The stronger the corporate culture for ASD projects, the more likely would be their success.

HO6a: The more informal the plans are in an agile project, the more likely would be the success of the ASD projects.

HO6b: The more quality control procedures agreed and administered by the project team, the more likely would be the success of the ASD projects. [Note: It should be clarified that it is not more quality controls per se that are the issue but quality control procedures agreed and administered by the project team rather than external quality assurance groups].

HP1: The more technically competent the team members are in a project, the more likely would be the success of the ASD projects.

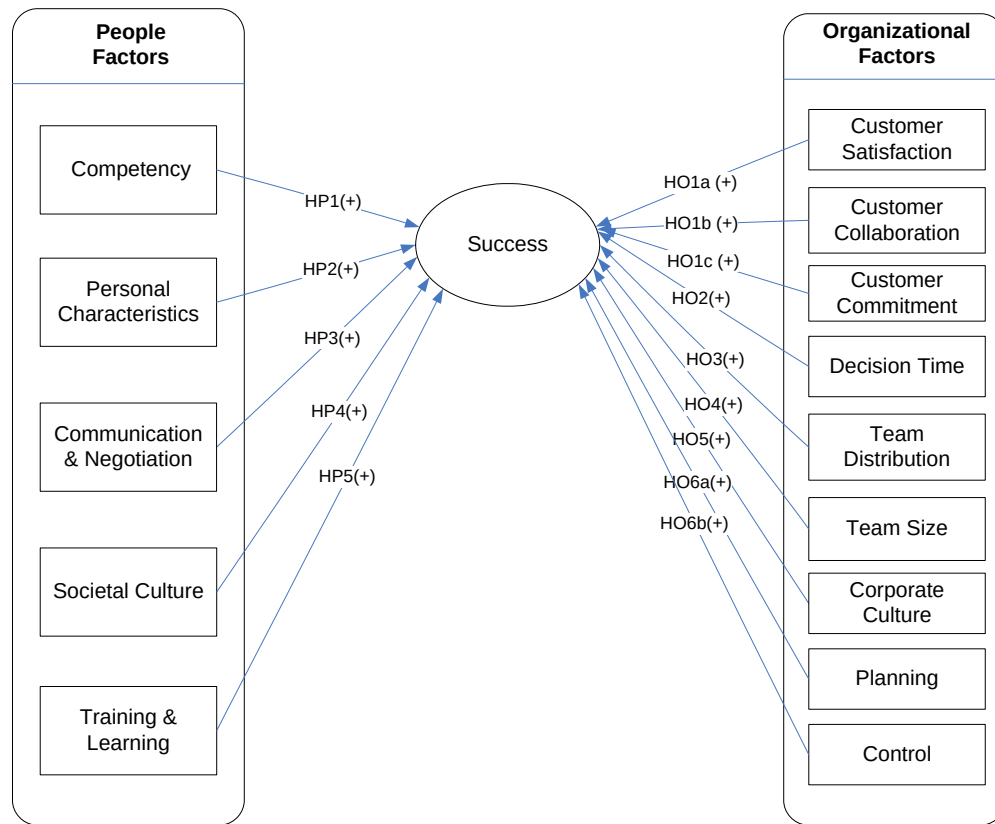


Fig. 1. Hypothesized success factors.

HP2: The better the personal characteristics of the team members in a project, the more likely would be the success of the ASD projects.

HP3: The more the communication and negotiation is between people, the more likely would be the success of the ASD projects.

HP4: The more favorable the societal culture is for the ASD projects, the more likely would be the success of the ASD projects.

HP5: The more the environment is for continuously learning, and informal training, the more evident would be the success of the ASD projects.

4. Research methodology

The research methodology we used in this work is one that is typically used in survey-based research in management information systems. The core of the methodology is survey-based, *post-facto* investigation. Broadly, it consists of the following three phases:

Phase I: Formulation of the research question, construction of a hypothetical framework (based on intuition and relevant literature) and the development of hypotheses based on the research question.

Phase II: Designing the survey instrument and the collection of data through surveys.

Phase III: Analysis of the data collected as part of Phase II.

We used the hypothetical framework we had constructed, and the hypotheses we had formulated, to design a questionnaire, which were *pre-tested* for its validity, usefulness, and readability before it was sent out for gathering data from the field. The data that were

gathered were then statistically analyzed using suitable data analysis techniques. We used those techniques to gain an understanding of the important factors responsible for the success of projects that want to adopt ASD practices. Multiple projects practicing ASD techniques were surveyed. We did not restrict our surveys to any particular industry type, sector, or size. The only requirement we had was that the survey respondents practice ASD, and they have transitioned in the past from traditional software development practices to those advocated by the ASD community.

The survey was administered using a *web-based* questionnaire, as shown in [Appendix A](#). Since we were interested in gathering the experiences of a large number of users of ASD practices, we sent our survey questionnaire to potential respondents in various industrial sectors (such as manufacturing, electronics, telecommunications, aerospace, oil and gas) that are practicing ASD techniques. Similarly, we did not restrict ourselves to any particular organization/project size, culture, or nationality. With this approach we were able to survey ASD professionals that are widely geographically distributed across continents. It was cost-efficient, and the respondents were able to answer the questions more thoughtfully by taking their own time. Additionally, the web-based questionnaire allowed the survey respondents to complete parts of the survey according to their own time. This also enabled them to do some “homework”, if necessary, before being able to answer some of the questions.

Respondents were asked to fill out a structured questionnaire (shown in [Appendix A](#)). The questionnaire has primarily closed-ended, multiple-choice type questions, with some open-ended questions at the end. The advantage of the open-ended questions is that they enable the respondents to provide some additional information, which was not captured in the close-ended questions. These open-ended questions were not used in statistical analysis, but the responses to such questions are summarized at the end

of Section 5. The multiple-choice type questions asked the respondents to rank their responses on a scale of 1 to 5 (Likert's scale). The close-ended multiple-choice type questions helped us perform statistical analysis on the available data.

In our research, identifying the respondents was a challenge, because ASD is still an emerging approach that has gained popularity primarily in the last 4–5 years. In organizations, often only a few teams practice ASD. Therefore, it was challenging to identify the specific respondents (and teams) in organizations who qualify to respond to our questionnaire. We did considerable amount of research on this issue, and have been fortunate to find the following sources for collection of data:

- CIO Magazine's August 2004 issue lists the awardee organizations of the "Agile 100" awards. However, this list identifies only the candidate organizations on which we can focus. Determining the specific contacts in these organizations still remained a challenging issue.
- Agile Alliance (<http://www.agilealliance.org>) has a publicly available list of all its corporate members. As they are members of the Agile Alliance, it is likely that these organizations have teams who practice ASD.
- Agile Alliance was also requested to circulate the questionnaire to its individual members as they were not willing to reveal the contact e-mail addresses of those members. They agreed to help in this process.
- Previous research papers and experience reports published in this area by industry-based authors were collected. Those authors were approached by e-mail to fill out the web-based questionnaire.
- The contact person of different agile user groups throughout the world was contacted by e-mail to fill out the web-based questionnaire. They were also requested to forward the questionnaire to other members in their user group and also to the other team members at their place of work.
- The committee members of the different agile conferences that are currently announced or have already taken place in the past were also approached.
- Several blog and website postings of the survey link were also done. The respondents were also requested to help in doing this in blogs they were associated with.

In almost all cases in which we were successful in obtaining the specific names of individuals who potentially practice ASD, we sent out a reminder after a week or so to remind them, in case they had not already filled out the questionnaire.

The depth of data analysis that we could perform depended on the number of responses that we could obtain. Our initial target was to obtain at least 150 responses. The respondent eligibility criteria that were set to consider responses for further consideration of data analysis were the following:

- (1) Does the respondent's team practice at least six of the twelve² ASD principles?
- (2) Has s/he ever been part of a team that used to practice traditional, plan-driven software development, and had adopted ASD practices in the past?

² The rationale behind this is that it was difficult to find large number of projects which follow all the 12 ASD principles. This is because as ASD is still an emerging philosophy, finding projects that follow ASD practices is difficult, and finding projects that follow all the practices was even more difficult (if not, nearly impossible). Therefore, initially we set the target to identifying projects that follow at least 50% of those principles. While there was no hard rationale behind choosing this figure, we felt that this figure was neither too high nor too low.

We received 241 responses, of which there were 174 eligible responses, i.e., responses which satisfied the eligibility criteria mentioned above, for the success factors survey.

It was necessary to pre-test the questionnaire, before sending it out for the actual survey. Request for pre-testing was sent out to ten potentially qualified software development professionals. Out of the 10 requests, only five agreed to help (50% turnaround rate). Pre-testing was finally done with the help of those five software development professionals. They had different experience levels and work functions and worked in both the government and the private sectors. While four of the five respondents did not have any major suggestions for improvement, one suggested a few changes for improving the clarity of the questions. Overall, there was not much revision that was required as a result of the pre-testing.

Pre-testing the questionnaire was helpful for clarifying any ambiguities in the questionnaire, its readability, and the ordering of the questions. Based on the results of the pre-test, the original questionnaire was revised to reflect the feedback of the pre-test respondents.

5. Analysis and results

The approach for data analysis was quantitative (statistical) in nature, with some qualitative analysis of the data collected through replies to the open-ended questions in the questionnaire. We performed different statistical studies with the processed data of the closed-ended questions. The variables and the corresponding labels used and the study are presented in Table 1. On the basis of the survey data we obtained, the summary, in percentage figures, of which is presented in Appendix B, we wanted to understand the success factors from the perspective of software practitioners in ASD projects. It was a relationship study, and we had 14 independent variables (some of which have multiple questions measuring them) and a consolidated dependent variable ("Success"), which has 5 parts in our study. In addition, we went one step further, and showed the regression models with each of these constituents of Success as a dependent variable and the rest of the 14 potential factors as the independent variables.

5.1. Test of reliability and validity

The consistency and the stability of a score from a measurement scale are assessed with reliability tests. Internal consistency analysis is usually performed to assess the reliability of a multiple-item reliability test. This helped us to understand how consistent the different questions (presented in Appendix A) are within a variable and how consistently the survey respondents responded to the items. We performed item-to-item correlation, item-to-scale correlation, the effect of deletion of an item on reliability. Some of the important results are presented below.

For testing internal consistency reliability, we used Cronbach's alpha.³ Cronbach's alpha helps in estimating the proportion of systematic or consistent variance in a given sample of test scores. The value of Cronbach's alpha lies between 0 (no consistent variance can be accounted for) and 1 (variance is all consistent). A higher value of Cronbach's alpha indicates a greater consistency in variance of the sample test scores. Usually, a Cronbach's alpha value greater than 0.6 is considered standard in survey research. Some statisticians use 0.7 as the required cut-off point. A higher value of Cronbach's alpha would indicate a greater reliability or accuracy of the statistical inferences from the data. For example, if in a study the

³ There is another variation of the commonly used Cronbach's alpha, called the "Standardized Item Cronbach's alpha". It is computed by standardizing the scale items so that they have equal means and variances. Some researchers use this value of alpha in their tests. In our study, we present the results of both of these versions.

Table 1
Variables used.

Variables	Label
Customer satisfaction	Ind1
Customer collaboration	Ind2
Customer commitment	Ind3
Decision time	Ind4
Team distribution	Ind5
Team size	Ind6
Corporate culture	Ind7
Planning	Ind8
Control	Ind9
Technical competency	Ind10
Personal characteristics	Ind11
Communication and negotiation	Ind12
Societal culture	Ind13
Training and learning	Ind14
Reduced delivery schedules	D1
Increased return on investment (ROI)	D2
Increased ability to meet with the current customer requirements	D3
Increased flexibility to meet with the changing customer requirements	D4
Improved business processes	D5
Success	Dependent

Table 2
Reliability statistics summary.

Items in	Number of items	Chronbach's alpha	Chronbach's alpha based on standardized items
Ind5	2	.726	.704
Ind7	8	.872	.878
Ind11	6	.847	.852
Ind12	5	.700	.664
Ind13	4	.784	.779
Dependent	5	.717	.744

Cronbach's alpha for a set of test scores turns out to be 0.7, it would mean that there is 70% consistency, and $100 - 70 = 30\%$ of inconsistency of test scores. Table 2 shows the Cronbach's alpha of those constructs that are measured with multiple items. As we can see, the Cronbach's alpha for all items are at greater than or equal to 0.6 (in fact, they are mostly at least 0.7). This indicates that there are no problems with the internal consistency reliability tests.

5.2. Descriptive statistics

The summary of the survey responses in percentage figures is presented in Appendix B. First we performed descriptive statistics to help us in summarizing the information latent in the data that we obtained from surveys for this research question. First, for each question corresponding to both the independent and the dependent variables,⁴ we tried to gain an understanding of the *minimum*, *maximum*, *mean*, and *standard deviation* of the survey data obtained from respondents. Table 3 shows the descriptive statistical results. It should be observed that, just by considering the mean values, amongst all the 14 independent variables Ind1 has the highest mean of 4.77 with a standard deviation of 0.59. In other words, most of the participants in this survey strongly agree that they give highest priority to customer satisfaction more than anything else. Ind2 (customer collaboration), Ind3 (customer commitment), Ind4 (decision time), Ind6 (team size), Ind7 (corporate culture), Ind10 (technical

Table 3
Summarized descriptive statistics.

	N	Mean	Std. deviation
<i>Descriptive statistics</i>			
Ind1	174	4.76	.595
Ind2	174	4.17	.925
Ind3	174	3.95	1.074
Ind4	173	4.30	.815
Ind5	174	3.22	1.267
Ind6	174	4.64	.899
Ind7	174	4.16	.793
Ind8	174	3.70	1.227
Ind9	174	3.60	1.501
Ind10	173	4.11	.985
Ind11	174	4.39	.577
Ind12	173	4.02	.748
Ind13	172	3.75	1.001
Ind14	171	4.34	.783
Dependent	173	4.37	.557
Valid N (list wise)	167		

competency), Ind11 (personal characteristics), Ind12 (communication and negotiation), and Ind14 (training and learning), according to the survey results were also considered to be in the range of (somewhat agree, strongly agree). The means of all the independent variables also show that Ind5 has the lowest mean value of 3.22. This shows that most of the participants remained neutral regarding whether most of the team members in their projects were geographically closely located, and the other teams that they interact with are geographically closely located. It should also be observed that none of the independent variables has a mean score below 3 ("Neither Agree nor Disagree" in the 5-point Likert scale that we used). This shows that on an average, the participants that were surveyed were not in disagreement with the factors in their projects.

In our survey, we also collected the background information of the respondents. The majority of the respondents (around 32.5%) belong to industries whose primary line of business is computer related, followed by respondents who belong to consulting organizations (around 29.8%). There were also around 10.2% responses from organizations having the primary line of business as banking/insurance industries, around 5.1% having the primary line of business as education/research, and around 2.4% each from medical/health organizations, and aerospace organizations. In addition to these organizations, there were about 7.9% respondents who belong to other organizations having primary line of business such as travel, logistics, transportation, government contractor, agriculture, and integrated workplace management. There were no responses from the hospitality, legal services, and non-profit organizations. Thus, we can see that there has been a fair share of responses from different organization sectors.

Of all the respondents, roughly 27.6% belong to organizations having 501–1000 employees, roughly 26.8% having more than 1000 employees, roughly 11.8% having less than 10 employees, roughly 9.8% having 21–40 employees, roughly 7.9% having 41–100 employees, and roughly 5.9% having 10–20 employees. Thus, we can see that the majority of the respondents belong to organizations having more than 500 employees. The number of responses from small-sized organizations is also roughly around 27–28%. Thus there has been fairly good response from both relatively small-sized and large-sized organizations.

Most of the responses (roughly 33.3%) came from respondents belonging to teams having strength of 5–10 employees, roughly 26.3% in teams having 11–20 employees, and roughly 19.4% in teams having less than 5 employees. Around 10.3% responses each were obtained from teams having 21–40 employees, and greater than 40 employees. Thus, we can see that more than 50% of the respondents belong to small-sized teams (having less than 10

⁴ Each of the dependent and the independent variables were given a label during statistical analysis. The different labels used are summarized in Table 1.

employees). Roughly another 50% responses came from teams having more than 10 employees. This shows a fair distribution of participation from all sizes of teams.

Most of the responses (about 29.5%) were obtained from developers/testers, about 18.9% from team leaders, about 17.7% from project managers, and 7.9% from functional managers. There were also about 25% responses from respondents having other roles such as director, business analyst, software architect, scrum master, and usability analyst.

The majority of the respondents (roughly 30%) have been developing software using agile principles/methods for 1–3 years, 26.4% for 3–5 years, 24% for greater than 5 years, and 19.6% for less than 1 year. This shows that we had participants from fairly all duration levels.

In order to assess the degree to which the different respondents practice ASD principles, the respondents were asked to mark their response in the 1 to 5 Likert scale (1 – Strongly Disagree, and 5 – Strongly Agree). The response average values imply that on an average the majority of the respondents agree with the statement that they practice most of the principles to a great extent (greater than 4). In fact, a majority of the respondents have stated that for all of 12 principles they strongly agree with the statement that they follow them.

5.3. Linear multiple regression analysis and correlation analysis

Correlation analysis helped us to understand the degree of relationship between all our 14 independent variables and the dependent variable (“Success”). Table 4 shows the summarized correlation results, between the dependent variable and each of the independent variables, for items with non-missing values.

This analysis results in mainly two sets of values: (i) the Pearson’s coefficient of correlation and (ii) the values of significance. In this study, the test was conducted at level 0.05. On the basis of the data that we have obtained, we see that the independent variables Ind1, Ind2, Ind3, Ind4, Ind7, Ind9, Ind11, Ind13, and Ind14 are statistically significantly related to the dependent variable, Success, whereas the independent variables Ind5, Ind6, Ind8, Ind10, and Ind12 were found not to have any significant relationship with Success. In sum, solely on the basis of correlation analysis, the following factors are significantly related to Success:

- Customer satisfaction
- Customer collaboration
- Customer commitment
- Decision time
- Corporate culture
- Control

Table 4
Summarized correlations table.

Variables	Correlation coefficient	Significance
Ind1	.140	.036
Ind2	.232	.001
Ind3	.253	.001
Ind4	.254	.000
Ind5	.074	.171
Ind6	.097	.107
Ind7	.241	.001
Ind8	.075	.168
Ind9	.301	.000
Ind10	.102	.096
Ind11	.180	.010
Ind12	.104	.092
Ind13	.381	.000
Ind14	.283	.000

- Personal characteristics
- Societal culture
- Training and learning

Solely on the basis of our survey results, the following were found *are not* significantly related with Success:

- Team distribution
- Team size
- Planning
- Technical competency
- Communication and negotiation

On the basis of the significance values and the values of correlation coefficients, we either *accept or reject our 14 hypotheses* we have listed in Section 3. The acceptance/rejection results of hypotheses mentioned in Section 3 are as shown below⁵:

HO1a: The greater the satisfaction of the customers in projects, the more likely would be the success of ASD projects. **Accepted.**

HO1b: The greater the collaboration of the customers in projects, the more likely would be the success of ASD projects. **Accepted.**

HO1c: The greater the commitment of the customers in projects, the more likely would be the success of ASD projects. **Accepted.**

HO2: The quicker the decisions are taken in projects, the more likely would be the success of ASD projects. **Accepted.**

HO3: The more closely located the project teams are, the more likely would be the success of ASD projects. **Not Accepted.**

HO4: The smaller the size of the teams in a project, the more likely would be the success of ASD projects. **Not Accepted.**

HO5: The stronger the corporate culture for ASD projects, the more likely would be their success. **Accepted.**

HO6a: The more informalized the plans are in an agile project, the more likely would be the success of ASD projects. **Not Accepted.**

HO6b: The more qualitative controls the projects have, the more likely would be the success of ASD projects. **Accepted.**

HP1: The more technically competent the team members are in a project, the more likely would be the success of ASD projects. **Not Accepted.**

HP2: The better the personal characteristics of the team members in a project, the more likely would be the success of ASD projects. **Accepted.**

HP3: The more the communication and negotiation is between people, the more likely would be the success of ASD projects. **Not Accepted.**

HP4: The more favorable the societal culture is for agile projects, the more likely would be the success of ASD projects. **Accepted.**

HP5: The more the environment is for continuously learning, and informal training, the more evident would be the success of ASD projects. **Accepted.**

The outcome of the multiple regression analysis that was conducted is a table containing the coefficients of the estimated regression model. It is shown in Table 5.

Tables 5 and 6 show the regression coefficients and the regression model summary that were obtained. The *R* values for each model estimate the coefficients of multiple correlation. It indicates the degree of linear relationship between the observed and pre-

⁵ It should be noted that instead of marking some of the hypothesis as “Rejected”, we have marked them as “Not Accepted” because such an evaluation is done based only on the statistical analysis that we have performed.

Table 5

Regression coefficients.

Model	Unstandardized coefficients		Standardized coefficients	t	Sig.	Correlations			Collinearity statistics	
	B	Std. error	Beta			Zero-order	Partial	Part	Tolerance	VIF
Coefficients ^a										
1 (Constant)	1.268	.039		32.860	.000					
Ind13	.052	.010	.381	5.274	.000	.381	.381	.381	1.000	1.000
2 (Constant)	1.131	.052		21.850	.000					
Ind13	.053	.010	.388	5.590	.000	.381	.401	.388	.999	1.001
Ind3	.034	.009	.264	3.801	.000	.253	.285	.264	.999	1.001
3 (Constant)	1.081	.052		20.686	.000					
Ind13	.049	.009	.358	5.275	.000	.381	.383	.355	.982	1.018
Ind3	.031	.009	.239	3.522	.001	.253	.267	.237	.987	1.013
Ind9	.021	.006	.230	3.369	.001	.301	.256	.227	.972	1.029
4 (Constant)	.940	.068		13.889	.000					
Ind13	.044	.009	.323	4.819	.000	.381	.355	.316	.955	1.047
Ind3	.031	.008	.246	3.728	.000	.253	.282	.244	.986	1.014
Ind9	.020	.006	.213	3.197	.002	.301	.244	.209	.966	1.036
Ind14	.037	.012	.212	3.179	.002	.283	.243	.208	.962	1.040

^a Dependent variable: log dependent.

dicted values of Success. As we can see, the values of *R* for all the above four models produced by the stepwise regression procedure belong to the range (0, 1). In general, greater the *R* values, the stronger the relationships are. The proportion of variation in Success explained by the regression model is estimated by the *R* squared values. The *R* squared values also lie in the range (0, 1). Model 4 has the highest value of *R* squared amongst all the four models. It has four predictors: Ind13, Ind3, Ind9, and Ind14. Thus, stepwise regression, in this example, reveals only four variables as predictors of Success. This is consistent with the commonly used idea that there should not be too many predictors in a good multiple regression model. In order to better reflect the goodness of fit of the model, the adjusted *R* squared is used. The adjusted *R* squared for Model 4 is also the highest amongst all the four models. In Table 5, the *B* values in the “unstandardized coefficients” column are the coefficients of the estimated regression model.

By looking at the *t*-statistics in Table 5, we can estimate the importance of one variable relative to another in the model. The general rule of thumb used for looking at *t* values is to look for *t* values below −2 or above +2. Here, all the variables Ind13, Ind3, Ind9, and Ind14 have *t* value well above +2. Also, the collinearity statistics indicates that in Model 4, all the VIF values are well below the threshold of 2 (VIF values greater than 2 are considered problematic), and all the tolerance values are above the threshold of 0. This indicates that with this model, multicollinearity is not a problem.

Table 7 summarizes the results of analysis of variance corresponding to each of the 4 models. Corresponding to each model, the sum of squares, degrees of freedom, and mean square are shown. The amount of variation that can be accounted for by each of these four models is shown in Regression row of the table, the

Table 6

Regression model summary.

Model	<i>R</i>	<i>R</i> square	Adjusted <i>R</i> square	Std. error of the estimate
1	.330(a)	.109	.104	.12685
2	.436(b)	.190	.180	.12136
3	.491(c)	.241	.227	.11783
4	.530(d)	.281	.263	.11505

a Predictors: (Constant), Ind13.

b Predictors: (Constant), Ind13, Ind3.

c Predictors: (Constant), Ind13, Ind3, Ind9.

d Predictors: (Constant), Ind13, Ind3, Ind9, Ind14.

e Dependent Variable: LOG (base e) DEP.

amount of variation that cannot be accounted for by each of these models is shown in the Residual row, and the total variation (i.e., the sum of the variations accounted for by both the Regression, and the Residual) is also shown thereafter for each of the four models. Amongst all the four models, Model 4 has the largest regression sum of squares compared to the residual sum of squares. It has the least residual compared to the other three models. The *F* statistic results in the table reveal that the independent variables Ind13, Ind3, Ind9, and Ind14 are good (significant) predictors of variation in Success, because the significance value (i.e., .000) is less than .05 (the test of significance cut-off value).

Ideally, for a good regression model, the residuals should follow a normal distribution. Fig. 2 shows the histogram of the regression standardized residual, which is the value of the difference between the observed value and that predicted by the model for the dependent variable, Success. As the shape of the histogram in Fig. 2 closely follows the shape of the normal curve, it is acceptable. Similarly, from Fig. 3 we see that the P–P plotted residuals follow the 45-degree line. Thus, we see that neither the histogram nor the P–P plot indicates that the normality assumption in linear multiple regression is violated. Fig. 4 shows that the error variance is more or less constant. We may mention that there exists a very

Table 7

ANOVA table from regression analysis.

Model		Sum of squares	<i>df</i>	Mean square	<i>F</i>	Sig.
ANOVA ^e						
1	Regression	.454	1	.454	27.815	.000 ^a
	Residual	2.679	164	.016		
	Total	3.133	165			
2	Regression	.672	2	.336	22.270	.000 ^b
	Residual	2.461	163	.015		
	Total	3.133	165			
3	Regression	.833	3	.278	19.573	.000 ^c
	Residual	2.299	162	.014		
	Total	3.133	165			
4	Regression	.969	4	.242	18.031	.000 ^d
	Residual	2.164	161	.013		
	Total	3.133	165			

^a Predictors: (Constant), Ind13.^b Predictors: (Constant), Ind13, Ind3.^c Predictors: (Constant), Ind13, Ind3, Ind9.^d Predictors: (Constant), Ind13, Ind3, Ind9, Ind14.^e Dependent variable: Log dependent.

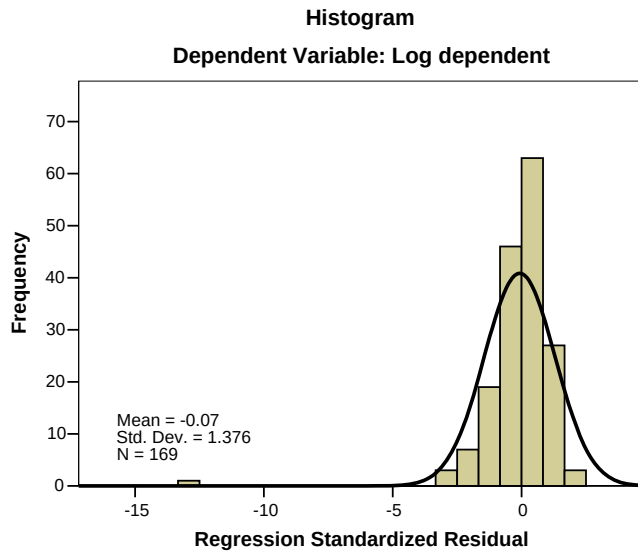


Fig. 2. Histogram showing frequency versus regression standardized residual.

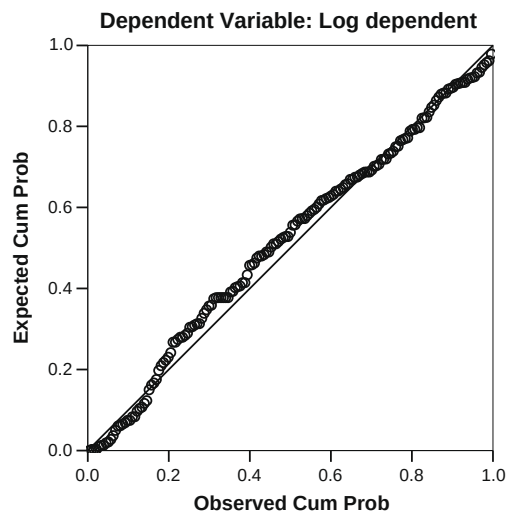


Fig. 3. Normal P-P plot of regression standardized residual.

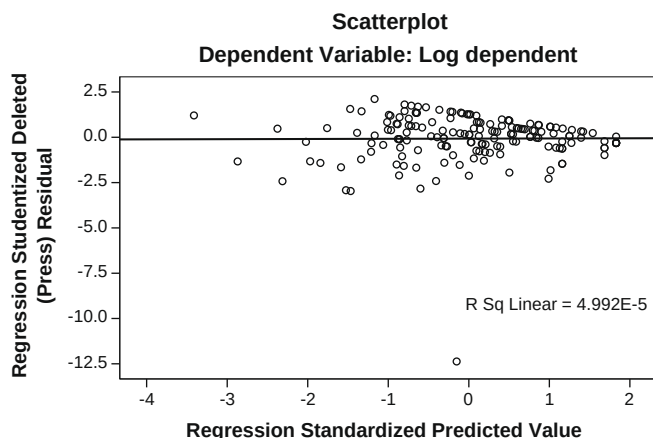


Fig. 4. Check for error variance.

Table 8

Regression Models for D1, D2, D3, D4, and D5.

Predictors	Regression model
1 Ind13, Ind14	$D1 = 2.343 + .255 (\text{Ind13}) + .197 (\text{Ind14})$
2 Ind14, Ind9, Ind2	$\ln (D2) = .626 + .098 (\text{Ind14}) + .048 (\text{Ind9}) + .052 (\text{Ind2})$
3 Ind3, Ind7	$\ln (D3) = 1.232 + .036 (\text{Ind3}) + .038 (\text{Ind7})$
4 Ind3	$D4 = 4.24 + .013 (\text{Ind3})$
5 Ind4, Ind7, Ind3	$\ln (D5) = .722 + .079 (\text{Ind4}) + .052 (\text{Ind7}) + .033 (\text{Ind3})$

large outlier and that we have checked that the results are resilient to the removal of the outlier.

In exactly similar manner, stepwise linear multiple regression was also run on the data taking each of the five dependent variables, D1, D2, D3, D4, and D5 individually as representatives of Success, instead of consolidating them all together to represent Success, as was done thus far. The regression models that are obtained for each of D1, D2, D3, D4, and D5 are presented in Table 8.

5.4. Success factors suggested by respondents (from open questions)

The respondents were also requested to list up to five *additional* success factors. As it was an optional question, some of the participants responded to it, while some did not. For maintain the brevity of this paper, the detailed open-ended responses of the respondents cannot be present here. A detailed inspection of all the suggested factors reveal the following high-level categories to which all the factors can be classified:

- Customer centric issues
- Learning from failure
- Training and learning
- Competency
- Personal characteristics
- Communication
- Timing issues
- Corporate culture and management issues
- Use of tools
- Team characteristics
- Planning
- Control

It should be noted that many of the success factor dimensions/categories that we have identified above, overlap with those of the ones that we had previously identified. Therefore, we list all the dimensions even if they were previously identified, so that we do not lose any data that we have obtained. However, it should be noted that what is important for us is the identification of the major dimensions of the success factors. The list of factors contributing to any of the dimensions that we have identified could be large (not restricted to the ones we had in the questionnaire and the ones brought forward through the open-ended questions).

6. Discussions

We have seen that solely from the analysis of the quantitative data obtained, nine factors emerge to have statistically significant relationship with Success, whereas five do not. The ones that emerge to be so are customer satisfaction, customer collaboration, customer commitment, decision time, corporate culture, personal characteristics, societal culture, and training and learning. Among the ones that did not show significant relationship are team distribution, team size, planning, technical competency, and communication and negotiation. Based on the citations from the existing

pieces of literature in this area, and the results obtained from the analysis of the qualitative responses obtained through the open-ended questions, all the nine factors that show statistically significant relationship with Success are strongly justifiable. However, after analyzing the results obtained corresponding to the other five factors that did not emerge as significant success factors, using logical justification, using support from the previously published literature, and using the responses obtained through the open-ended questions, we feel that there are in-depth studies that should be performed to establish or reject the observed results through this study. Let us revisit all the hypothesized factors and analyze the findings from different perspectives.

6.1. Customer centric issues

It is worth noting that all the three hypothesized customer centric issues viz., customer satisfaction, customer collaboration, and customer commitment were found to have a significant relationship with Success.

Although solely from the correlation significance point of view all of these three customer centric issues were found to have a significant relationship, by looking at their coefficients of correlation it can be seen that customer commitment was found to have a strongest relationship of all, followed by the customer collaboration, and customer satisfaction respectively. This relative behavior is observed solely based on the analysis of the data obtained, and there is no apparent logical justification or literature support that can be provided in support of or against this behavior. It is worthy of further investigation using other means. In an absolute sense, customer commitment and customer collaboration are undoubtedly very important from the point of view of the agile practitioners. It is, however, interesting to observe that while customer collaboration and customer commitment were at least somewhat agreed to be practiced in approximately 78% and 71% percent of cases, customer satisfaction was at least somewhat agreed to be practiced in approximately 97% of the cases. This is, again, not surprising because irrespective of what approach the project teams take, it is likely that most people would agree that they try to satisfy their customers.

Viewing it from what the different respondents have suggested as other success factors in the open-ended questions, almost all of them suggested in different forms that customer commitment, and customer collaboration were important success factors. There were no significant additional suggestions in favor of the customer commitment factors. Thus the results of the open-ended questions are also intrinsically suggestive of the customer commitment and collaboration in different morphs. Development teams want the customers to remain closely associated, engaged, and committed to the project. They do not want buying software to be viewed by the customers as similar to buying commodities such as groceries! Thus the results of the analysis of the suggestions obtained from the open-ended questions are also in conformance with those obtained from the analysis of the data obtained from the closed-ended question.

The whole idea of agility is trying to develop software efficiently so as to satisfy the customers. This requires that the customers are committed on the project, and they involve/engage themselves closely with the development team. It is worth recalling that customer commitment was even found to be one of the four predictors of the consolidated dependent variable Success in its regression model measured with the help of the five measures mentioned earlier in the paper, whereas customer collaboration was found to be a predictor of the increased return on investment taken alone as a measure of Success.

In conclusion, it can be said that it is sensible that we found significant relationship between Success and customer centric issues.

However, it is contextual to warn that there may be practical challenges trying to engage customers. In an interesting article (Deias et al., 2002), the authors describe their experiences of introducing the agile methodology XP in a start-up company, and found that there were several challenges from the customers while involving customers and collaborating with them. For example, they mention:

“Our customers do not like contractual provisions that address the risk of changing requirements. Most of our customers regard software as just one of the goods that they acquire, and want to buy it using the same kind of contractual agreements used for buying other products. . . . The fear of the business people is that by opening up the development process to our customers, we will end up having customer going around shopping for specific people, trying to build the team of their choice for specific projects. . . .”

There are also other practical challenges of having customers on-site that have been brought forward by them, that should be addressed.

Jensen (2002) reports his experience from several projects that for having success in a project, cooperation between users (customers) and developers is very important. The importance of customer commitment, collaboration, and involvement is also evident from the findings of Graffin (2001) from implementing an agile methodology in an organization. Our findings regarding the importance of such customer centric issues as important success factors is further strengthened by the findings from such pieces of literature.

6.2. Decision time

The factor, decision time, also demonstrated significant correlation with Success. The coefficient of correlation also shows that the strength of the relationship is relatively high as compared to the rest of the 14 factors. These results are not surprising considering that the whole idea behind agility is being fast. Fast communication, informal communication, effective communication, which are cited by many researchers such as, Turner and Boehm (2003) and Ambler (2005a) in different ways enable fast decision making (Lindvall et al., 2002), which in turn leads to the success of projects practicing agile software development. This is further evidenced from the fact that 84% of the agile practitioners that were surveyed agreed that they strove to make important project decisions rapidly and within short timeframes. Timing is an important factor. As seen from the additional success factor responses obtained from the survey participants, cutting down on the time to delivery and time to market are suggested as important factors. One of the respondents even considered “time boxing” as one of the most important factors. It is suggested that to be successful, it is important to scale down the requirements, and slice the whole time into short durations. The implicit idea behind this is that by doing all of the above, among other advantages, the decisions can be made more rapidly and efficiently. By looking at the responses to the open-ended questions, we feel that a better construct in the hypothesized theoretical framework could have been the consolidated factor “timing issues”. However, we based all the constructs in the theoretical framework initially purely based on the surveyed literature, and came up with the “decision time”, which captures a singular dimension of the timing issues.

6.3. Corporate culture

The factor, corporate culture, was also found to be statistically significantly correlated with Success. The strength of this relationship was also found out to be high relative to the strengths of the other 14 factors that were considered in this study.

It is not surprising to see that the responses from the practitioners of agile software development have shown significant relationship with Success, considering the citations from the existing pieces of literature. For instance, Lindvall et al. (2002) have mentioned that:

“To be agile is a cultural thing. If the culture is not right, then the organization cannot be agile”.

Many experts agree that the corporate culture is an important success factor (Lindvall et al., 2002). It may be recalled that successful agile software development requires that the organizations are dynamic and are fast changing (Abrahamsson et al., 2002). The organization should have the right culture such as supporting rapid communication, dynamicity in requirements changes, trusting people, and obtaining fast feedback from customers. These were evidenced from the discussions in Lindvall et al. (2002).

Our results are as well in conformance with the findings of Bossavit (2002) in using an agile methodology in two different organizations that there are, indeed, certain cultural values that are necessary. The following statements of the author provides good light into the issue:

“...the most crucial feature of any successful corporate culture must be a willingness to reflect upon itself, to assess whether it needs to change itself, and, if need be, to embark on such changes as it finds necessary”.

Corporate culture is an important success factor, indeed. In fact, the statements made by Bossavit (2002) not only reflect upon corporate culture, but also the importance of different personal characteristics. This is not surprising because organizations are made up of people and the characteristics of people are undoubtedly very important to success.

The analysis of the percentage responses obtained against each of these eight measures of corporate culture we considered in this study suggests that the majority of the practitioners of agile software development strongly agree that they have all the above cultural factors in their organizations.

It should be further observed that although corporate culture has not been found during regression analysis to be a significant predictor of the consolidated dependent variable, Success, it was found to be a predictor of the Success measures, increased ability to meet with the current customer requirements, and improved business processes, each taken in isolation as the dependent variable.

Corporate culture, as a success factor, was further evidenced from the responses of the open-ended success factor questions. Many of the additional success factors that were put forward by the different respondents broadly captures the dimension of corporate culture. For instance, the respondents felt that there should be an environment of supportive management, there should be buy-in from all stakeholders (inclusive of management), there should be lack of politics, good customer–vendor relationship in the organization, and adaptability to changes.

6.4. Control

The factor, control, was found to have a statistically significant relationship with Success. By looking at the correlation coefficient, which indicates its strength of its relationship with Success, it is perhaps the second strongest (the strongest being societal culture, to be discussed shortly) amongst all the 14 hypothesized factors. We measured this construct by asking respondents whether they practice qualitative control (as against having quantitative performance measures). Of the majority of the respondents, approximately 62%, at least somewhat agreed to practice it.

Anderson (2002) reports a practical experience story with using “morning roll call” (daily standup) in having project success. Such a report supports our finding that qualitative control is an important success factor. The observations from statistical analysis are also supportive of the comments made in literature such as Boehm and Turner (2003), Cockburn (2002a,b), Cohn and Ford (2003) and McMahon (2004). Their observations and comments are also implicitly or explicitly suggestive of qualitative control.

Not only from the correlation analysis results, but also by observing the regression model for the consolidated dependent variable, Success, we see that it emerges as one of the four predictors of Success. However, in the regression models corresponding to each of the five dependent variables taken in isolation, it appears as a predictor only for increased return on investment, and it does not appear as a predictor for reduced delivery schedules, increased ability to meet with the current customer requirements, increased flexibility to meet with the changing customer requirements, and improved business processes. This indicates that control has a relatively strong influence on increasing the return on investment. Such a behavior is only observed from analyzing the data obtained from the closed-ended questions. Our literature survey cannot justify this observation either in favor or against. It is worthy of validation through further studies.

Analysis of the responses obtained through the open-ended success factor questions also suggest different qualitative ways of demonstrating project progress such as weekly progress show-cases to the customers, and/or daily standups. Interestingly, one of the respondents also suggested showing progress in terms of “completed story points per iteration”.

In sum, as has been evidenced from statistical analysis, qualitative analysis of the open-ended responses, and the existing literature, qualitative control emerged to be a significant success factor.

6.5. Personal characteristics

Although personal characteristics were also found to have a significant relationship with Success, the strength of its relationship relative to the other hypothesized factors is not commendable. This observation is somewhat surprising when the existing literature is revisited.

It is often felt that good people are key to success with big teams (Lindvall et al., 2002). Personal characteristics such as having honesty, collaborative attitude, sense of responsibility, readiness to learn, and work with others are often felt as important success factors (Lindvall et al., 2002). Technical expertise and professional qualifications alone cannot succeed a project. The importance of personal characteristics such as cooperativeness is also reported in the literature (Jensen, 2002). Similar observations suggestive of the above facts can be made by reviewing the following pieces of literature: Ambler (2005b), Boehm and Turner (2003, 2005); Cockburn and Highsmith (2001) and Cockburn (2002a,b).

The majority of the respondents at least somewhat agreed (in fact, most of them strongly agreed) to have team members having such kind of personal characteristics. However, it did not show a strong relationship with Success compared to the other factors. However, a number of responses to the open-ended success factor questions were suggestive of personal characteristics as an important success factor. They are as follows: Having intellect; Taking up responsibility; Having a base of good principles; Willingness of programmers to work as pairs; Passionate; Not being a perfectionist; Self-criticism; Courage to tell the truth about project; Respect of individual; Good People; and Humility.

Although it is not to be denied that the construct personal characteristics do show that its relationship with success is significant, we expected that the relationship would be stronger than what we

obtained. This observation is contrary to earlier research that is supportive of the role of personal characteristics. Therefore, further studies using other methodologies should be warranted to closely observe this observed phenomenon.

6.6. Societal culture

The consolidated percentage figures for the responses through the four measures of societal culture we considered in this study suggest that a majority of the surveyed team members have somewhat agreed to possess or abide by the above dimensions of societal culture.

Societal culture was observed to have a significant relationship with Success suggesting that having people in the society who are communicative, dynamic, progressive in attitude, and having team members with similar social culture, taken together, are important success factors. The correlation analysis results also demonstrate a relatively very strong coefficient of correlation (in fact, the strongest of all), thereby being strongly suggestive of the presumption based on observations from the existing literature that societal culture does play a significant role as a success factor.

It should be noted that multiple regression analysis suggests that societal culture was found to be a significant predictor of the consolidated Success variable amongst all the 14 factors considered together. Even when regression models were constructed with each of the five measures of Success taken in isolation, it was observed that societal culture is a significant predictor of the return on investment. This observation is obtained purely from the data obtained from the closed-ended questions, and cannot be supported with the existing literature that were surveyed.

Interestingly, the additional success factor responses obtained from the respondents against the open-ended questions, do not reveal any additional dimensions of societal culture.

6.7. Training and learning

The construct, training and learning, was found to have a statistically significant correlation with Success. Furthermore, it was also found to have a relatively strong relationship. These results are not surprising because there is a strong evidence in the existing literature (Boehm and Turner, 2005; Drobka et al., 2004; Lindvall et al., 2002) which is suggestive of training and learning as a success factor.

This construct was measured with the determination of the willingness to continuously learn from one another and train the team members through mentoring and professionally guided discussions than through formal training. Thus the emphasis is on continuous, informal learning. The results showing that the majority of the respondents strongly agree that they practice this approach of training and learning in their teams. The results showing significant and relatively strong correlation with Success imply that the majority of the agile practitioners that were surveyed believed that continuous, informal learning (over formal training) was a significant success factor. Informal learning helps in knowledge diffusion in an organization. However, it should be noted that there are authors such as Dingsoyr and Hansen (2004) who believe that informal learning methodologies such as pair programming alone cannot be effective means of knowledge diffusion in an organization. Dingsoyr and Hansen (2004) have suggested postmortem reviews as a lightweight mechanism to aid in such a purpose.

The regression model for Success also shows training and learning to be a significant predictor of Success. Even the results of the regression modeling with each of the five measures of Success, taken in isolation as a dependent variable, also show the training and learning construct to be a significant predictor of the reduced delivery schedules and increased return on investments.

The above results are further supported by the responses to the open-ended questions, which indicate usage of informal training methods such as pair programming. This will enable transfer of knowledge through discussion of designs. However, there were a few respondents who believed in having full time coaches and taking help of outside coaching services to succeed.

6.8. Team distribution

The results of the analysis of the closed-ended data obtained from the respondents do not suggest any significant relationship with Success. While the majority of the respondents maintained that they strongly agreed to have geographically closely located members of their own team, they strongly disagreed with the need to have other teams with which they interact geographically closely located to theirs. Thus, from the results of the correlation analysis it can be suggested that geographical collocation of own team members or members of other teams with which interactions are made does not show any significant relationship with Success.

The above results are contrary to the suggestions from existing literature for having collocated teams as a success factor (e.g., Lindvall et al., 2002). Further, there are earlier authors (Ambler 2005a; Boehm and Turner, 2003; Cockburn, 2001; McMahon, 2004; Turner and Boehm, 2003) who have emphasized on close proximity, and communication issues such as face-to-face communication. Our findings are also not in conformance with the suggestions made by Turk et al. (2002), in which they have mentioned the identification of limited support for distributed development of agile processes as one of the important factors in agile development. In other words, team distribution should be an important success factor. In this context, it is worth citing the experience of Motorola, Inc., in their global development projects. Copeland (2001) mentions that Motorola, Inc. had used an agile methodology, XP, in some of their development projects, and found that it was not suitable for their global development projects. The reason cited relates to the fact that XP encourages working in small teams, and they had difficulty doing so in their global development projects. This is further suggestive of the fact that team distribution should be an important factor in determining success.

The apparent inconsistencies with the literature that we have obtained through the analysis of the data obtained from the closed-ended questions could be attributed to the differences in measures of success that we considered in this study, with the ones that are suggested by the authors. The suggestions to the open-ended questions from the respondents are not suggestive of any significant success factors that can be attributed to this dimension. This behavior could be partially justified by the fact that may be a significant number of respondents would not justify team distribution, or even communication and negotiation, as a major success factor as compared to many of the others that we considered. Let us take the example of the case reported by Fowler (2006) in which he reported success in using agile practices in offshore development project, in spite of the fact that in offshore development environments communication and negotiation would pose major challenges. Fowler (2006) mentioned some strategies which could successfully overcome the challenge of having distributed teams.

Thus, because it is hard to justify the results we have obtained, team distribution is a candidate for further detailed investigation using other research methodologies such as case studies.

6.9. Team size

The construct, team size, failed to demonstrate a significant correlation with Success. Using this construct it was measured whether the respondents worked in small teams. This is contrary to the popular belief in the literature that we surveyed. Ambler

(2005a), among others, emphasize on close face-to-face communication. Having small teams was suggested in the literature (Dyba, 2000b). There are suggestions in the literature (Lindvall et al., 2002), about what constitutes a small team. Typically, it is less than 4–25 members. The rejection of small team size could be connected with these suggestions because we have surveyed different types of people working in different project types.

The reason, however, why we hypothesized small team size to be a success factor is, logically, and as mentioned in some of the above mentioned pieces of literature (see the discussions on team size in the Lindvall et al. (2002)), having a small team size enables effective communication, fast feedback, and prompt decision making.

Our findings are inconsistent with the Turk et al. (2002) who have identified that limited support for development involving large teams in agile processes as one of the challenges in agile development. Also Parrish et al. (2004) in their early study found that productivity influences of pair programming as the team/organization size increases. Since pair programming is a common practice in agile software development, this gives us another clue that team size should be an important factor.

Of course there are other instances reported in the literature of using agile practices in large team environments. Pine (2001) mentions his positive experience of successfully using XP (a methodology following the agile practices) in a large, multi-team environment. Of course, whether he received success according to our definition cannot be inferred from his paper. The instance of Harrison (2003) is also worth noting. They have reported of positive experiences using agile practices in large projects that have large number of teams, and large team size. All these stories of different opinions may also partially help us justify our observation from data analysis that team size or project size may not be a success factor.

Because of the varying thoughts, opinions, arguments, and justifications regarding this issue, we call for further research to investigate this issue in detail. In particular, we suggest that different projects with varying team size should be investigated in detail preferably using a case study based approach, which might be helpful in making detailed observations regarding this issue.

6.10. Planning

Planning was not found to have a significant relationship with Success. The aspect of planning that we measured is internalized, informal, and undocumented plans (as against formal plans). Insignificant relationship with Success is contrary with the literature we surveyed which suggest internalized planning over the formal ones (Boehm and Turner, 2003; Cockburn, 2002a,b; Cohn and Ford, 2003; McMahon, 2004). This indicates that formal planning cannot be discarded from consideration. Although, by abiding by the literature, we hypothesized for internalized undocumented planning, and against formal planning, it being not a success factor is not counter-intuitive. Agile practices are relatively new, most experienced people we surveyed have been working in traditional type projects for years, and it is unlikely that by surveying such kind of people we will get its strong relationship with Success. Furthermore, many of these people work in complex projects, involving a large number of personnel, and in big organizations, which would require documented planning. Since there are multiple factors that affect Success, intuitively, there can be Success in a project (attributable to other factors), even when formal planning was undertaken. The above statements could possibly justify the results we have obtained through data analysis of responses to the closed-ended questions for the possibility of “planning” as a success factor.

6.11. Technical competency

It was found out through analysis of data from closed-ended questions that technical competency does not have a statistically significant correlation with Success. The majority of the people that we surveyed at least somewhat agreed that they have technically competent people. However, no significant relationship was found between technical competency and Success. First, this is counter-intuitive. This is because, having competent or incompetent people should make a difference to the success of a project. Secondly, our literature survey (Cohn and Ford, 2003; Boehm and Turner, 2005; Cockburn and Highsmith, 2001) is also implicitly or explicitly suggestive of the competency dimension. Also, this is contrary to Barry Boehm’s suggested principle of using fewer and better people (Boehm, 1981).

In regards to competency as a success factor, the practical experience of Deias et al. (2002) introducing agile in a start-up Internet company is also worth noting. According to them:

“If the team lacks the necessary programming experience results will be at best marginally better than what one would expect from any other methodology”.

Our initial hypothesis of having competency as a success factor is also not in conformance with the observations of Deias et al. (2002). Experience and competency are very important.

In addition to the citations from the literature, when the respondents were surveyed through open-ended questions for additional success factors, there were a number of them who suggested different factors that are suggestive of different competencies such as the following: Talent; Developers know the business and can sometimes be their own customers; Smart people; Individual capabilities of team members; Craftsmanship; Developers’ experience; Domain knowledge; Developer familiarity with the technical platform; and Developer skills.

In view of the above discussions, we are unable to come to any conclusions regarding the insignificant relationship of the competency construct with success. We see that many respondents have made many suggestions in favor of different technical and non-technical competencies. Thus, we call upon further investigation regarding this issue.

6.12. Communication and negotiation

Communication and negotiation were also found not to have significant relationship with Success, based on the survey data obtained through the closed-ended questions. This is, perhaps, the most surprising of all the above factors that were found not to have a significant relationship with Success.

Several previous pieces of literature support effective communication mechanisms to be in place for success. Some of them that are implicitly or explicitly suggestive of the same include Ambler (2005a), Cockburn (2001), Lindvall et al. (2002), Turner and Boehm (2003), and McMahon (2004). Fast and effective communication supports fast decision making, adapting with the dynamicity of projects, and fast knowledge transfer between team members in a project.

Considering the points of view raised through the additional open-ended success factor questions, as well, we see that a number of factors such as open and continuous feedback between team members, using simple but effective communication tools, using information radiators such as charts and cards on the walls, and using communication rules that are suggestive of communication as a success factor were raised.

7. Limitations and scope

The studies of this sort are mostly accompanied by different limitations of varying severities. It is, therefore, important to view the research outcome presented in this paper within the bounds of these limitations. These limitations and the ones specifically presented in Section 8 should be considered together to inspire future research work on this topic.

1. In this work, we attempted to find statistical evidence for the success factors using a survey mechanism. However, such a technique has its own limitations. For instance, it is not advisable to construe these findings as universally applicable in all kinds of environments such as project types and sizes, application domains and organizational and communal cultural domains.
2. The variables used in the study are each complex in their nature. There are highly subjective and there is likely to be only limited conceptual agreement within the field. Therefore, the results relating to them should be viewed accordingly. For instance, corporate culture was measured in terms of eight factors. The definition of corporate culture in this work is a function of these factors only. Thus, all results reported in this paper that relate to corporate culture relate to only these dimensions of this construct.
3. Due to lack of sufficiently related literature in the domain area, the pieces of literature that are used to conceptualize each construct are not always scientific. This may sometimes influence the research findings. We tried to measure each of these variables in terms of multiple factors. We had to resort to measuring each of the constructs in terms of only a selection of constituent factors. The more the number of such constituent factors, the better representative the construct measures what it is trying to measure. However, too many such factors measuring a construct have a negative impact on the length of the survey instrument. In our case, we had to restrict ourselves to only a few such constituent factors, as we were dealing with a large number of constructs in the survey instrument already.
4. The methodology of asking a few people for their opinion without a mediated focus group methodology has its limitations, even if the questions were tested to be reliable in terms of Chronbach alpha and the questionnaire was pre-tested for its validity, usefulness, and readability before it was sent out for gathering data from the field. Although this research methodology is often undertaken in similar kinds of research in empirical software engineering and management information systems, it is possible that biasness and other psychological factors may have an influential effect on the overall research results. It is, therefore, advisable to treat the research results accordingly. More fresh and follow-up work in the future are called for to address some of these deficiencies.
5. In this work, based on existing literature and intuitive reasoning, we hypothesized a conceptual model of success factors and then used quantitative data to validate/invalidate the different components of the model. It is important to recognize that the results presented in this paper are limited within the bounds of the initially hypothesized success factors. Although we believe we have covered a broad spectrum of success factors, it is possible that by adopting other research methodologies, other potential success factors may emerge. This is partially why we recognize this in the title of this paper itself by stating that we are only attempting to identify *some* important success factors.
6. In this study, although we do not have a *truly* random sample from a defined population (i.e., an actual list of all population members), the inclusions/exclusion criteria for individuals seems appropriate for a non-statistical survey. However, a non-statistical survey does have generalization problems. For example, it is unlikely that our search strategy would encourage anyone without a personal commitment to agile methods to complete the survey (i.e., any individuals who tried agile and planned developments and reverted to planned developments would be unlikely to participate). However, the goal of the study requires us to obtain the responses from the perspective specific of the agile practitioners.
7. In this work, we have limited ourselves to analyzing the consolidated data obtained from various respondents of different continents around the world. We have not conducted a detailed investigation of the transcontinental/cultural differences in the success factors. As we may have realized from our previous discussions, ASD is influenced by behavioral/personal characteristics of respondents. Such an investigation would require capturing not only the country/continent of origin of the respondent, but also their places of work. There is a significant amount of complexity involved in this kind of research, because in today's global economy, organizations are distributed throughout the world, many employees travel across continents, they might have been educated/trained in one culture, but their work behavior has changed over the years by working in different cultural environments. The investigation of this issue was scoped out of this study.
8. Similar to the point mentioned above, whether the project size, project complexity, or the organization size matters or not was scoped out from investigation of this work. Therefore, the research results are limited within this scope. While the value of such a work was appreciable, it was felt that inclusion of such a research question would increase the length of the already long questionnaire (it should be noted that many people have commented critically about the size of the questionnaire). However, in this work, we collected such information to get a statistic of the size/complexity of the organizations/projects of the respondents of the survey. This was done with the idea that these statistics would hopefully help us to get an insight about the type of respondents of the survey whose responses have produced the results that we have obtained.
9. We targeted individuals but asked them about team information (including other liaising teams) and company issues and indeed societal issues. This means that responses from individuals in the same team, company or the same country would not be independent for some questions. In cases where the questionnaire was circulated through user groups and other forums, it was indeed not possible to identify how many of the respondents belonged to a particular company, project or society. We maintain that, this is, in fact, something that is best achievable using other research methodologies such as case studies and interviews. Other similar issues, such as the relationship between societal-based questions and the nationality of respondents and checking whether respondents from different countries are consistent with respect to the societal-based questions are certain issues that have been left behind. This would require us to capture other information such as the nationality of the respondents. As the questionnaire did not capture the ethnic origin, the nationality of

the respondents, etc., we list this issue here as a limitation of this study worth investigating in future studies of similar nature.

10. Tests of validity are important in studies of this nature. Although we performed detail regression analysis and Chronbach's reliability tests, we did not plan properly to perform different validity tests. Particularly, content validity, criterion-related validity and construct validity are performed in studies of this nature. Content-validity is helpful to understand the nature of effect of the scale items on the domain under study. Criterion-related validity measures how the scales under investigation relate with an independent measure of the relevant criterion. Construct validity examines how the measurement scale is representative of the attributes measured. Data collection and analysis results show that although we did not explicitly perform these tests, the results we obtained largely, although not completely, conform to the requirements of these tests of validity and their expected results. However, unless we can thoroughly design from the beginning for performing such tests and actually perform them, we should not make firm comments concerning validity.
11. This study is limited within the assumption bounds that the data obtained across different work functions are equally important. It would have been quite interesting to investigate if at all there will be any differences in the results based on the work functions of the respondents. However, that required changes in the original survey design and instrument preparation – we leave this as a future activity of potential interest.
12. One of the issues got unnoticed during the survey instrument design. In the case of team distribution, the idea that individuals are “neutral” about whether their company has co-located or distributed teams is not appropriate and is more like to be an indication of the fallacies that occur when ordinal scale measures are treated as if they were interval or ratio scale. It is far more like that some individuals work in collocated teams and others do not. This would be indicated by a bi-modal distribution of the basic responses which should not be represented by a means and standard deviation and is probably inappropriate for correlation analysis. Future studies should take this into account. However, even in the current design, a bi-modal response will be indicated by a respondent as one of the extremities in the ordinal scale. A intermediate response would indicate that the team is not extremely co-located or extremely distributed, i.e., there are mixed degrees of practices existing in the team.

7.1. Future work

Many issues are worthy of detailed investigation in the future. Two important ones, in addition to the ones that are evident from addressing the limitations outlined in Section 7.1, are outlined below:

1. Further research should be done to validate the results obtained from this study in practice. Practitioners should attempt to use the results obtained in this study in actual projects and test the degree of usefulness of the results in practice. This is important because although the results obtained in this paper have been obtained by statistically analyzing the responses from currently practicing agile software professionals, unless they are deemed

to be of use in practice, the research results would be of limited use other than serving as an inspiration for future researchers to conduct further research in this direction.

2. Due to the nature of these studies, controlled experiments, in which one factor is kept constant while varying others to different degrees, are difficult to conduct in real-life projects. To overcome this, it is suggested that computer-based simulation studies should be performed to validate the results obtained from this study. Such controlled experiments would be helpful to examine the detailed behavior/characteristics of any particular issue.

8. Conclusions

The overall goal of this work was to advance the state-of-the-art of the research in this area by identifying factors that lead to the success of software development projects that want to adopt ASD practices, from the perspective of ASD practitioners. We undertook a survey-based ex-post-facto empirical (quantitative) study approach by identifying the success factors from the perspective of software practitioners in ASD projects. First, an hypothetical framework was developed. Thereafter, different software development practitioners, practicing agile principles in their projects, were approached with both closed-ended and open-ended type questions. While the closed-ended questions formed the core of the study, the open-ended questions helped in getting fresh perspectives from currently practicing ASD practitioners from different industry sectors. The responses were obtained from different industry sectors including those having the primary line of business as computer related (hardware/software/IS/MIS), consulting, banking/insurance, healthcare, and aerospace. The respondents belonged to both small and large-sized organizations and teams. Responses came from all role types – software developers/testers, and all management levels.

We conducted an unprecedented large-scale empirical study. Most of the previous empirical studies have been localized within a few projects/organizations. The feedback we received while conducting the survey has been overwhelming. Many of the respondents, who are also experts in this area, have shown interest in the work and its results.

Our research *contribution* can be summarized as follows:

- Determining the factors that lead to the success of software development projects that want to adopt ASD practices. We have found that 9 of the 14 hypothesized factors significantly relate with success. They are customer satisfaction, customer collaboration, customer commitment, decision time, corporate culture, control, personal characteristics, societal culture, and training and learning. Team distribution, team size, planning, technical competency and communication and negation were not found to have a significant relationship with success.
- We also arrived at linear multiple regression models for the consolidated dependent variable success, and the regression models for five measures of success we considered in the study viz., reduced delivery schedule, increased return on investment, increased ability to meet with current customer requirements, increased flexibility to meet with changing customer requirements, and improved business processes.
- Determining additional success factors, if any. From the responses obtained from the open-ended questions, we were successful in finding additional success factors. Among others,

four seemingly important success factor dimensions have emerged. They are: learning from failure, timing issues, other team characteristics, and use of tools. The above items did not initially appear in our hypothesized list of success factors. This is because we hypothesized all the success factors based on our search of existing literature, and our search of existing literature did not find sufficient evidences supporting the consideration of such factors in the literature. This, however, does not indicate that these newly suggested success factors are not

important. As the identification of such success factors is a complex problem guided by several factors, we call for additional research to examine them.

Appendix A. Survey questionnaire

The checklist below will help in assessing the degree of practice of agile software development principles by your team. It contains

Response (Mark 'X')	Choices
<i>Respondent's Background Information</i>	
Primary <i>identity</i> of our organization	Computer related (IS/MIS/DP/Hardware/Software) Telecommunications Banking/insurance Real estate Business supplies/services Education/research Entertainment/media/publishing Hospitality Medical/health care Government Engineering/construction Consulting Legal services Manufacturing/distribution Consumer retail/wholesale Non-profit/membership organization Electrical machines Aerospace Other. (Please specify:_____)
The number of employees in our <i>organization</i> is	Less than 10 10–20 21–40 41–100 101–500 501–1000 Greater than 1000
The number of employees in our <i>team</i> is	Less than 5 5–10 11–20 21–40 Greater than 40
My <i>role</i> in the team:	Functional Manager Project Manager Team Leader Developer/Tester Other. (Please specify:_____)
I have become <i>knowledgeable</i> about agile software development practices for:	Less than 1 year 1–3 years 3–5 years Greater than 5 years
I have been <i>developing software</i> using agile principles/methods for:	Less than 1 year 1–3 years 3–5 years Greater than 5 years

12 statements. For each of the principles, please mark in the scale of 1 (disagree) to 5 (strongly agree) your degree of practice of each of the principles.

ment practices in your team within your organization. Please mark your choice on a scale of 1 to 5, where 1 is “Strongly Disagree”, and 5 is “Strongly Agree”. Your replies should reflect

We practice this principle	Response					
	Strongly Disagree	Somewhat Disagree	Neither Disagree nor Agree	Somewhat Agree	Strongly Agree	Not Applicable or Don't know (X)
	1	2	3	4	5	
<i>Checklist</i>						
1 We give high priority to satisfying customers through early and continuous delivery of valuable software						X
2 We welcome changing requirements, even late during development						X
3 We deliver working software more frequently, from couple of weeks to a couple of months, with a preference to a shorter timescale						X
4 Our business people and developers work together daily (very closely) throughout the project						X
5 We build projects around motivated individuals. We give them the environment and support they need, and trust them to get the job done						X
6 We emphasize more on face-to-face communication for conveying information to and within the development team						X
7 We measure and track progress based on working software						X
8 We promote sustainable development. Our sponsors, developers, and users maintain a constant pace indefinitely						X
9 Our software development project team follows continuous attention to technical excellence and good design for development						X
10 We practice simple designs, processes, and approaches in our software development methodologies. We implement features that are required by the customers – nothing more						X
11 Our development teams are self-organizing – our teams can (re)-organize continuously in different configurations to meet the changing requirements and the newly arising challenges of the business						X
12 At regular intervals, our team reflects on how to become more effective, then tunes and adjusts its behavior accordingly						X

A.1. Success factors survey

Please indicate your level of agreement of the following statements based on your experience of agile software develop-

your best guess from your knowledge and experience of software development projects performed by your team in your organization.

	Strongly Disagree	Somewhat Disagree	Neither Disagree nor Agree	Somewhat Agree	Strongly Agree	Not Applicable or Don't
<i>Customer satisfaction</i>						
In our projects, we give very high priority to achieving customer satisfaction	1	2	3	4	5	X
<i>Customer collaboration</i>						
In our projects, customers closely collaborate with the development team members	1	2	3	4	5	X

Appendix A.1 (continued)

	Strongly Disagree	Somewhat Disagree	Neither Disagree nor Agree	Somewhat Agree	Strongly Agree	Not Applicable or Don't
<i>Customer commitment</i>						
In our software development projects, customers are committed to the project, i.e., they are motivated, active, and consider themselves to be responsible elements of the project	1	2	3	4	5	X
<i>Decision time</i>						
We strive to make important project decisions rapidly within short timeframes	1	2	3	4	5	X
<i>Team distribution</i>						
The members in <i>our team</i> are geographically closely located	1	2	3	4	5	X
The <i>other teams</i> that we interact with within or outside our organization are geographically closely located to ours	1	2	3	4	5	X
<i>Team size</i>						
We work in small teams (no more than 20–40 members) in our projects	1	2	3	4	5	X
<i>Corporate culture</i>						
Our organization encourages rapid communication	1	2	3	4	5	X
Our organization has the culture for trusting people	1	2	3	4	5	X
Our management has the culture for supporting the decisions of the developers	1	2	3	4	5	X
Our organizational culture is customer centric	1	2	3	4	5	X
Our organization encourages fast feedback from customers	1	2	3	4	5	X
Our organization encourages changing requirements	1	2	3	4	5	X
Our organization does not have a bureaucratic management structure	1	2	3	4	5	X
Our management, developers, and testers are in total agreement to use agile practices	1	2	3	4	5	X
<i>Planning</i>						
Our software development team relies on internalized, informal, undocumented plans (as against formal documented plans)	1	2	3	4	5	X
<i>Control</i>						
Our software development team has qualitative control (as against quantitative performance measures)	1	2	3	4	5	X
<i>Technical competency</i>						
Our team generally consists of technically competent and experienced people (who have developed alike software in the past, have practical experience of the technology domain)	1	2	3	4	5	X
<i>Personal characteristics</i>						
The majority of our team members have strong <i>interpersonal and communication skills</i>	1	2	3	4	5	X
The majority of the members of our team consists of people who are <i>honest</i>	1	2	3	4	5	X
The majority of the members of our team consists of people who are <i>motivated</i>	1	2	3	4	5	X
The majority of the members of our team consists of people who have <i>collaborative attitude</i>	1	2	3	4	5	X
The majority of the members of our team consists of people who have <i>sense of responsibility</i>	1	2	3	4	5	X
The majority of the members of our team consists of people who have <i>readiness to learn</i>	1	2	3	4	5	X
<i>Communication and Negotiation</i>						
Our projects have mechanisms that enable personnel to communicate and negotiate <i>quickly and effectively</i> with developers, operations, support, customers, management, and business areas	1	2	3	4	5	X

(continued on next page)

Appendix A.1 (continued)

	Strongly Disagree	Somewhat Disagree	Neither Disagree nor Agree	Somewhat Agree	Strongly Agree	Not Applicable or Don't
In most cases, communication and negotiation in our projects are <i>face-to-face</i>	1	2	3	4	5	X
In most cases, communication and negotiation in our projects happen between people who are <i>physically close</i> to one another	1	2	3	4	5	X
In most cases, communication and negotiation in our projects happen between people who work in the same (<i>similar</i>) <i>time zone</i> as ours	1	2	3	4	5	X
Most people in our projects are amicable to each other to such an extent that they communicate with each other with <i>trust and good will</i>	1	2	3	4	5	X
<i>Societal culture</i>						
The people of our country are in general <i>communicative</i>	1	2	3	4	5	X
The people of our country are in general <i>dynamic</i>	1	2	3	4	5	X
The people of our country in general have <i>progressive attitude</i>	1	2	3	4	5	X
The majority of the members of our team have <i>similar social culture</i> , even though they might be belonging to different nationalities and provinces	1	2	3	4	5	X
<i>Training and learning</i>						
Our team members are in general always willing to continuously learn from one another and train them through mentoring and professionally guided discussions than through formal training	1	2	3	4	5	X

You are invited to specify any additional factors that you believe impact the success of agile software development projects. Please rate your responses on a scale of 1 to 5.

Other factors	Strongly Believe	Somewhat Believe	Neutral	Don't Strongly Disbelieve	Strongly Disbelieve
1	1	2	3	4	5
2	1	2	3	4	5
3	1	2	3	4	5
4	1	2	3	4	5
5	1	2	3	4	5

Please rate, on a scale of 1 to 5, your assessment of the degree of impact of practicing agile software development in your projects.

	Strongly Disagree	Somewhat Disagree	Neither Disagree nor Agree	Somewhat Agree	Strongly Agree	Not Applicable or Don't know (X)
Reduced delivery schedules	1	2	3	4	5	X
Increased return on investment (ROI)	1	2	3	4	5	X
Increased ability to meet with the current customer requirements	1	2	3	4	5	X
Increased flexibility to meet with the changing customer requirements	1	2	3	4	5	X
Improved business processes	1	2	3	4	5	X

Appendix B. Summary of survey responses

	1 (%)	2 (%)	3 (%)	4 (%)	5 (%)	X (%)
Ind1	0	1	1	20	77	1
Ind2	2	9	10	40	38	1
Ind3	3	10	14	38	33	1
Ind4	0	3	12	38	46	1
Ind5						
Ind5a	14	15	5	17	49	0
Ind5b	28	23	11	17	15	5
Ind6	3	2	4	11	79	1
Ind7						
Ind7a	1	6	7	22	63	0
Ind7b	2	7	9	25	56	0
Ind7c	4	7	8	33	47	1
Ind7d	1	7	14	35	42	1
Ind7e	4	10	7	27	50	1
Ind7f	6	13	15	31	34	0
Ind7g	10	12	8	22	47	0
Ind7h	4	14	12	29	38	2
Ind8	9	14	15	35	26	1
Ind9	4	10	16	31	31	8
Ind10	1	9	11	38	41	0
Ind11						
Ind11a	1	11	15	46	27	0
Ind11b	0	1	6	25	68	0
Ind11c	0	2	7	39	50	0
Ind11d	1	4	10	38	47	0
Ind11e	1	3	6	32	57	1
Ind11f	1	6	7	26	59	0
Ind12						
Ind12a	5	6	12	37	38	1
Ind12b	6	13	9	31	39	1
Ind12c	7	16	9	31	36	1
Ind12d	6	15	8	24	46	1
Ind12e	2	2	10	41	45	0
Ind13						
Ind13a	1	9	14	41	31	4
Ind13b	0	11	13	43	28	5
Ind13c	1	8	18	36	30	6
Ind13d	2	10	14	36	35	2
Ind14	2	3	11	39	45	0
Dependent						
D1	1	6	13	37	38	4
D2	1	3	12	30	47	6
D3	0	2	7	24	64	2
D4	0	2	4	20	71	1
D5	2	5	19	32	38	4

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